

Problem 1: Smart Railway Reservation System

Problem Statement

A railway reservation system stores the booking status of seats in a train coach.

Sample Data

```
seats = {  
    1: "Booked",  
    2: "Available",  
    3: "Booked",  
    4: "Available",  
    5: "Booked",  
    6: "Booked",  
    7: "Available",  
    8: "Booked",  
    9: "Available",  
    10: "Booked"  
}
```

Tasks

1. Display all available seat numbers.
2. Count booked and available seats.
3. Reserve the first available seat.
4. Cancel booking for a given seat number.
5. Store the updated reservation status in reservations.txt.
6. Display occupancy percentage.

Sample Output

Available Seats:
2 4 7 9

Booked Seats: 6
Available Seats: 4

Seat 2 Reserved Successfully.

Occupancy Percentage: 70.0%

Reservation Details Saved Successfully.

Problem 2: Hospital Patient Record Management System

Problem Statement

A hospital maintains patient details in a file named patients.txt.

Sample Input/Data (patients.txt)

P101,Anuj,Normal
P102,Rahul,Critical

P103,Priya,Stable
P104,Neha,Critical
P105,Amit,Stable
P106,Sneha,Normal
P107,Karan,Critical
P108,Pooja,Stable
P109,Rohit,Normal
P110,Anjali,Stable

Tasks

1. Display all patient records.
2. Display critical patients.
3. Count patients under each status.
4. Search patient details using Patient ID.
5. Save critical patient records to critical_patients.txt.

Sample Output

Critical Patients:

Rahul
Neha
Karan

Patient Count:

Normal : 3
Stable : 4
Critical : 3

Patient Found:

P104,Neha,Critical

Critical Patient Report Generated Successfully.

Problem 3: Smart Parking Management System

Problem Statement

The parking status of vehicles in a mall is maintained as follows.

Sample Data

```
parking_slots = [  
    "Occupied", "Vacant", "Occupied", "Vacant",  
    "Occupied", "Occupied", "Vacant", "Occupied",  
    "Vacant", "Occupied"  
]
```

Tasks

1. Display vacant parking slot numbers.
2. Count occupied and vacant slots.

3. Allocate the first vacant slot to a new vehicle.
4. Calculate parking occupancy percentage.
5. Store updated parking information in parking.txt.

Sample Output

Vacant Parking Slots:

2 4 7 9

Occupied Slots: 6

Vacant Slots: 4

Vehicle Allocated to Slot 2

Occupancy Percentage: 70.0%

Parking Details Saved Successfully.

Problem 4: School Report Card Generator

Problem Statement

Student marks are stored in marks.txt.

Sample Input/Data (marks.txt)

S101,Anuj,92
S102,Rahul,76
S103,Priya,88
S104,Neha,45
S105,Amit,58
S106,Sneha,95
S107,Karan,81
S108,Pooja,73
S109,Rohit,39
S110,Anjali,90

Tasks

1. Calculate grades for all students.
2. Generate a report card file report_card.txt.
3. Display topper details.
4. Count pass and fail students.
5. Display students eligible for merit certificates (marks ≥ 90).

Sample Output

Topper:

Sneha (95)

Passed Students: 9

Failed Students: 1

Merit Certificate Holders:

Anuj
Sneha
Anjali

Report Cards Generated Successfully.

Problem 5: Online Shopping Cart Analyzer

Problem Statement

The prices of products added to a shopping cart are stored below.

Sample Data

cart = [1500, 899, 450, 2500, 799, 1200, 300, 650, 1800, 999]

Tasks

1. Calculate the total cart value.
2. Find the most expensive and cheapest products.
3. Count products eligible for premium shipping (price > ₹1000).
4. Generate a discount list (products above ₹1500).
5. Calculate the average product price.

Sample Output

Total Cart Value: ₹11,097

Most Expensive Product: ₹2,500

Cheapest Product: ₹300

Premium Shipping Eligible Products: 4

Discount Eligible Products:
[2500, 1800]

Average Product Price: ₹1,109.7

Problem 6: Employee Attendance Monitoring System

Problem Statement

Employee attendance records are stored in attendance.txt.

Sample Input/Data (attendance.txt)

EMP101,P
EMP102,A
EMP103,P
EMP104,P

EMP105,A
EMP106,P
EMP107,P
EMP108,A
EMP109,P
EMP110,P

Tasks

1. Count present and absent employees.
2. Display absent employee IDs.
3. Calculate attendance percentage.
4. Generate an absentee report in absent_report.txt.
5. Display employees eligible for attendance awards (100% attendance).

Sample Output

Present Employees: 7

Absent Employees: 3

Absent Employee IDs:

EMP102
EMP105
EMP108

Attendance Percentage: 70.0%

Absentee Report Generated Successfully.

Attendance Award Eligibility:

Not Applicable

Problem 7: Movie Ticket Booking System

Problem Statement

Seat booking status in a cinema hall is stored as follows.

Sample Data

```
tickets = {  
    "A1": "Booked",  
    "A2": "Available",  
    "A3": "Booked",  
    "A4": "Available",  
    "B1": "Booked",  
    "B2": "Available",  
    "B3": "Booked",  
    "B4": "Available",  
    "C1": "Booked",  
    "C2": "Available"  
}
```

Tasks

1. Display available seats.
2. Count booked and available seats.
3. Reserve the first available seat.
4. Save updated booking details to tickets.txt.
5. Calculate hall occupancy percentage.

Sample Output

Available Seats:

A2
A4
B2
B4
C2

Booked Seats: 5

Available Seats: 5

Seat A2 Reserved Successfully.

Hall Occupancy Percentage: 60.0%

Booking Details Saved Successfully.

Problem 8: Smart Water Consumption Monitoring System

Problem Statement

Monthly water consumption (in litres) of households is recorded below.

Sample Data

```
water_usage = {  
    "House101": 1800,  
    "House102": 2200,  
    "House103": 3500,  
    "House104": 2800,  
    "House105": 1600,  
    "House106": 4100,  
    "House107": 2400,  
    "House108": 3900,  
    "House109": 1500,  
    "House110": 4500  
}
```

Tasks

1. Display houses consuming more than 3000 litres.
2. Find the highest and lowest consumers.
3. Calculate total water consumption.

4. Categorize houses:

- Low (<2000 litres)
- Medium (2000–3500 litres)
- High (>3500 litres)

5. Count households eligible for conservation awareness programs (>2500 litres).

Sample Output

Houses Consuming More Than 3000 Litres:

House103

House106

House108

House110

Highest Consumption:

House110 (4500 litres)

Lowest Consumption:

House109 (1500 litres)

Total Consumption: 28,300 litres

Low Consumption:

['House101', 'House105', 'House109']

Medium Consumption:

['House102', 'House103', 'House104', 'House107']

High Consumption:

['House106', 'House108', 'House110']

Eligible Households: 5

Problem 9: University Course Enrollment Management System

Problem Statement

Student enrollment data for university courses is stored below.

Sample Data

```
enrollment = {  
    "Python": 45,  
    "Java": 38,  
    "Data Science": 52,  
    "Web Development": 34,  
    "Machine Learning": 41,  
    "Cloud Computing": 29,  
    "Cyber Security": 33,  
    "DBMS": 48,  
    "Networking": 26,
```

```
"Operating Systems": 37
}
```

Tasks

1. Display courses having more than 40 enrollments.
2. Find the most and least popular courses.
3. Calculate total enrollments.
4. Create lists:
 - High Demand (>40)
 - Medium Demand (30–40)
 - Low Demand (<30)
5. Count courses requiring promotional activities (<35 enrollments).

Sample Output

Courses with More Than 40 Enrollments:

Python
Data Science
Machine Learning
DBMS

Most Popular Course:

Data Science (52 students)

Least Popular Course:

Networking (26 students)

Total Enrollments: 383

High Demand:

['Python', 'Data Science', 'Machine Learning', 'DBMS']

Medium Demand:

['Java', 'Web Development', 'Cyber Security', 'Operating Systems']

Low Demand:

['Cloud Computing', 'Networking']

Courses Requiring Promotion: 3

Problem 10: Smart City Waste Collection Management System

Problem Statement

The amount of waste collected (in kilograms) from different sectors of a city is stored below.

Sample Data

```
waste = {
    "Sector1": 320,
```



```
"Sector2": 180,  
"Sector3": 510,  
"Sector4": 275,  
"Sector5": 150,  
"Sector6": 430,  
"Sector7": 220,  
"Sector8": 390,  
"Sector9": 145,  
"Sector10": 600  
}
```

Tasks

1. Display sectors generating more than 400 kg of waste.
2. Find the sector generating maximum waste.
3. Find the sector generating minimum waste.
4. Calculate the total waste collected.
5. Categorize sectors:
 - Low Waste (<200 kg)
 - Medium Waste (200–400 kg)
 - High Waste (>400 kg)
6. Count sectors requiring awareness campaigns (waste generation >300 kg).
7. Save the awareness campaign list to campaign_sectors.txt.

Sample Output

Sectors Generating More Than 400 kg Waste:

Sector3

Sector6

Sector10

Maximum Waste Generation:

Sector10 (600 kg)

Minimum Waste Generation:

Sector9 (145 kg)

Total Waste Collected: 3220 kg

Low Waste:

['Sector2', 'Sector5', 'Sector9']

Medium Waste:

['Sector1', 'Sector4', 'Sector7', 'Sector8']

High Waste:

['Sector3', 'Sector6', 'Sector10']

Sectors Requiring Awareness Campaign:

Sector1
Sector3
Sector6
Sector8
Sector10

Campaign Report Generated Successfully.